

The Chemistry of Food

08/06/2026

Key Words: Sugar, protein, lipids, nutrients, reagent, starch, benedicts, iodine, biuret

DO NOW: Answer these three review questions below.

EXT: Which picture is the odd one out give a reason.



1. Name the 6 digestive organs. [3]

2. Give the journey of bile. Where it is made to where it is released. [3]

3. Bile is needed to neutralise stomach acid. Give another role of bile. [2]

1. Name the 6 digestive organs

**Mouth, oesophagus [1]
stomach, liver, [1]
Small and large intestine [1]**

2. Give the journey of bile

Liver [1] → Gall bladder [1] → small intestine [1]



ODD ON OUT
Food doesn't
travel through
the liver.

3. Bile is needed to neutralise stomach acid. Give another role of bile.

**It emulsifies fats into small droplets [1]
This increases the surface area so fats
Can be digested and absorbed faster [1]**

LOb: Understand the chemistry of food and how they can be tested.

Keywords: Sugar, protein, lipids, nutrients, reagent, starch, benedicts, iodine, biuret

Success Criteria:

- Recall the structure and role of the main nutrients
- Describe how to test foods for starch, sugar, lipids and protein
- Describe the positive result for each type of test



Iodine
53

126.90

Did you know... iodine is in group 7 of the periodic table. It has a lot of scientific uses. 1) Mixed with alcohol it can disinfect wounds 2) It is mixed with salt sometimes to prevent thyroid disease 3) Used as a catalyst.

Why do we do science tests on the food we eat?

To make the food more healthy

To make sure there is enough nutrients inside

To see if they taste nice for eating

To add nutrients to the food

To calculate amounts of nutrients

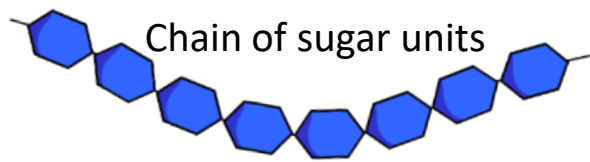
To make more money for businesses

To know what nutrients foods have in them

So we can tell what food it is.

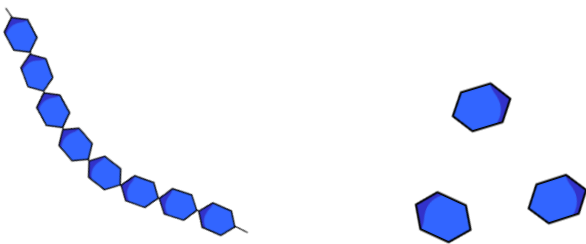


Nutrients in our food

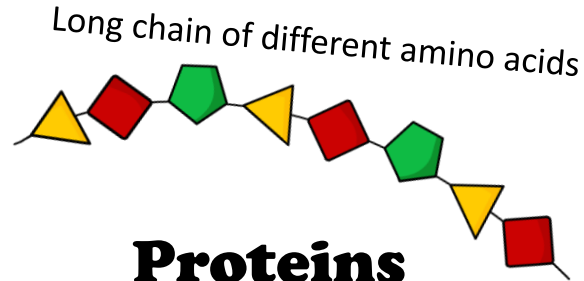


Carbohydrates

Provide fuel for the respiration

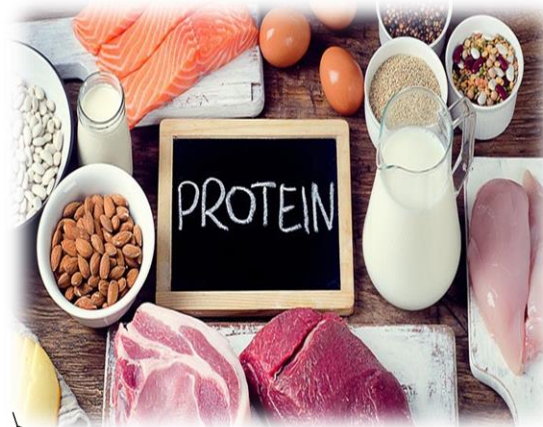


Starch → Glucose

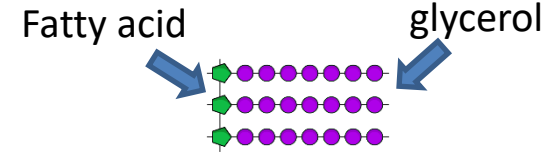


Proteins

Used to make cells/tissues structures and enzymes

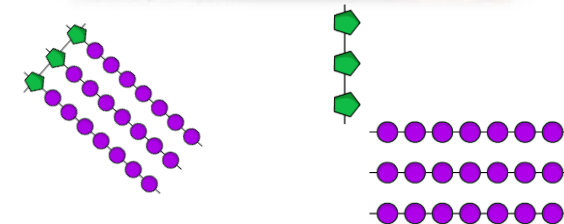


Protein → Amino acids



Lipids

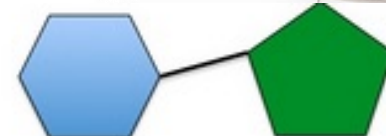
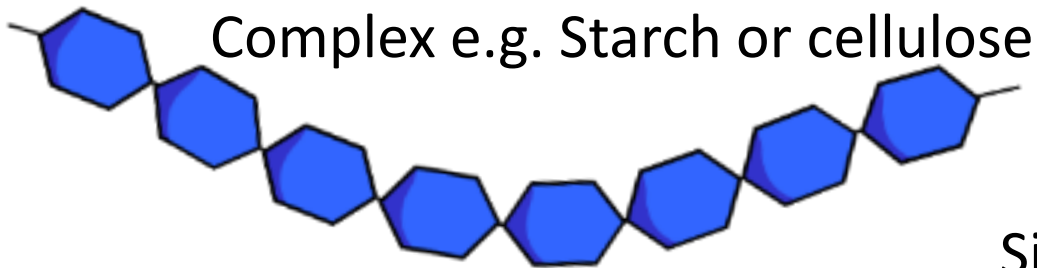
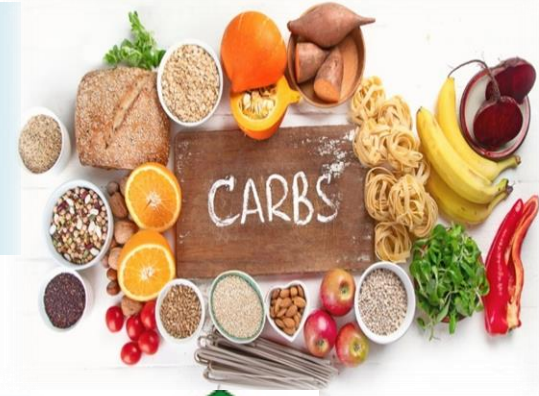
Efficient fuel source. Used to make hormones and Cell membranes



Fats → Fatty acids

Carbohydrates – Rice, potatoes, pasta etc.

Most carbohydrates **are broken down into glucose ($C_6H_{12}O_6$)** used in cellular respiration to **provide energy** for metabolic reactions in the cell.



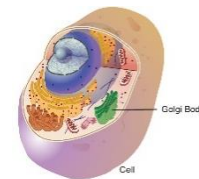
Simple e.g. Glucose or sucrose



Organised into simple and complex

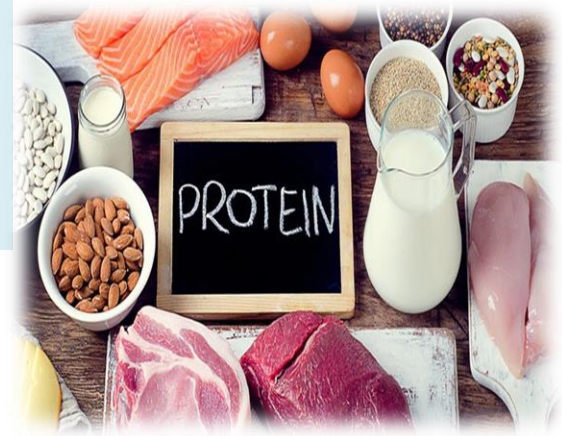


Broken down and used to release energy



Protein – meat, fish, eggs etc.

Protein is very important for life. In order to **make a protein you need amino acids** (found in our food). There are 20 different amino acids.



Proteins make up the structures of cells and tissues



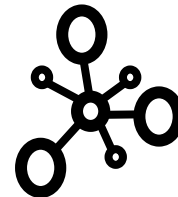
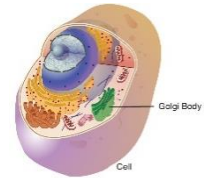
Proteins are used to make hormones like insulin



Proteins are used to make antibodies to kill pathogens

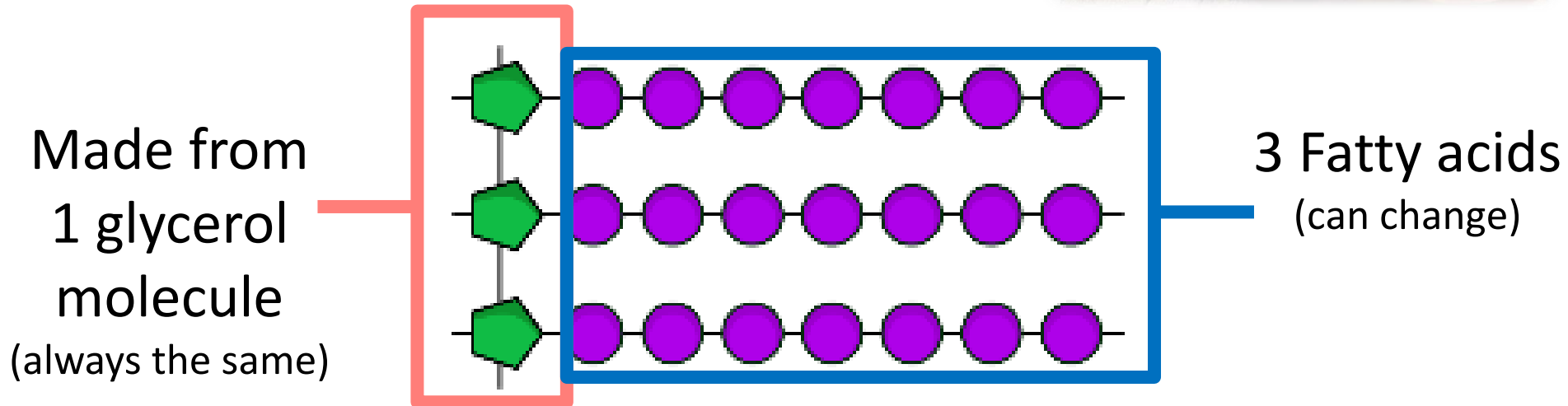


Enzymes are made of proteins



Lipids – Butter, cheese, cream etc.

Lipids are fats (solids) or oils (liquids). They are all **insoluble in water**. They are made from the same elements as carbohydrates.



**Efficient
energy source**

**Combined with other molecules to
make cell membranes and hormones**

LOb: Understand the chemistry of food and how they can be tested.

Keywords: Sugar, protein, lipids, nutrients, food test, starch, benedicts, iodine, biuret

Success Criteria:

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Iodine
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Did you know... iodine is in group 7 of the periodic table. It has a lot of scientific uses. 1) Mixed with alcohol it can disinfect wounds 2) It is mixed with salt sometimes to prevent thyroid disease 3) Used as a catalyst.

Preparing a food test sample

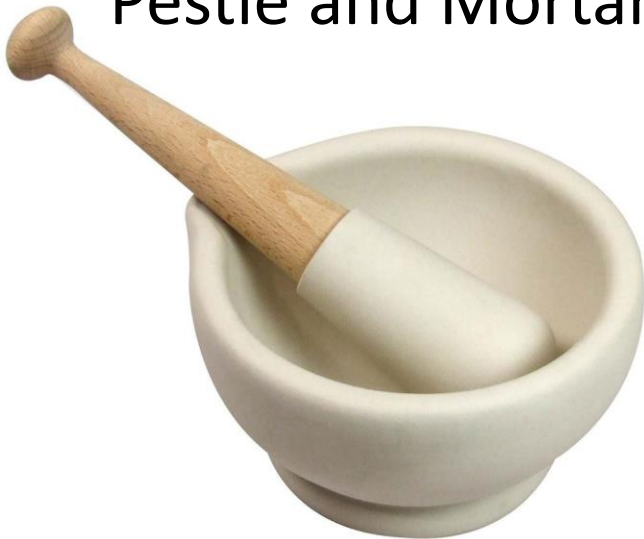
You need to know the test done for starch (carbohydrate), protein and sugar.

We used a substances called **reagents**.

Creating a sample of food:

1. Crush the food you want to test with a pestle and mortar
2. Add distilled water and mix well

Pestle and Mortar

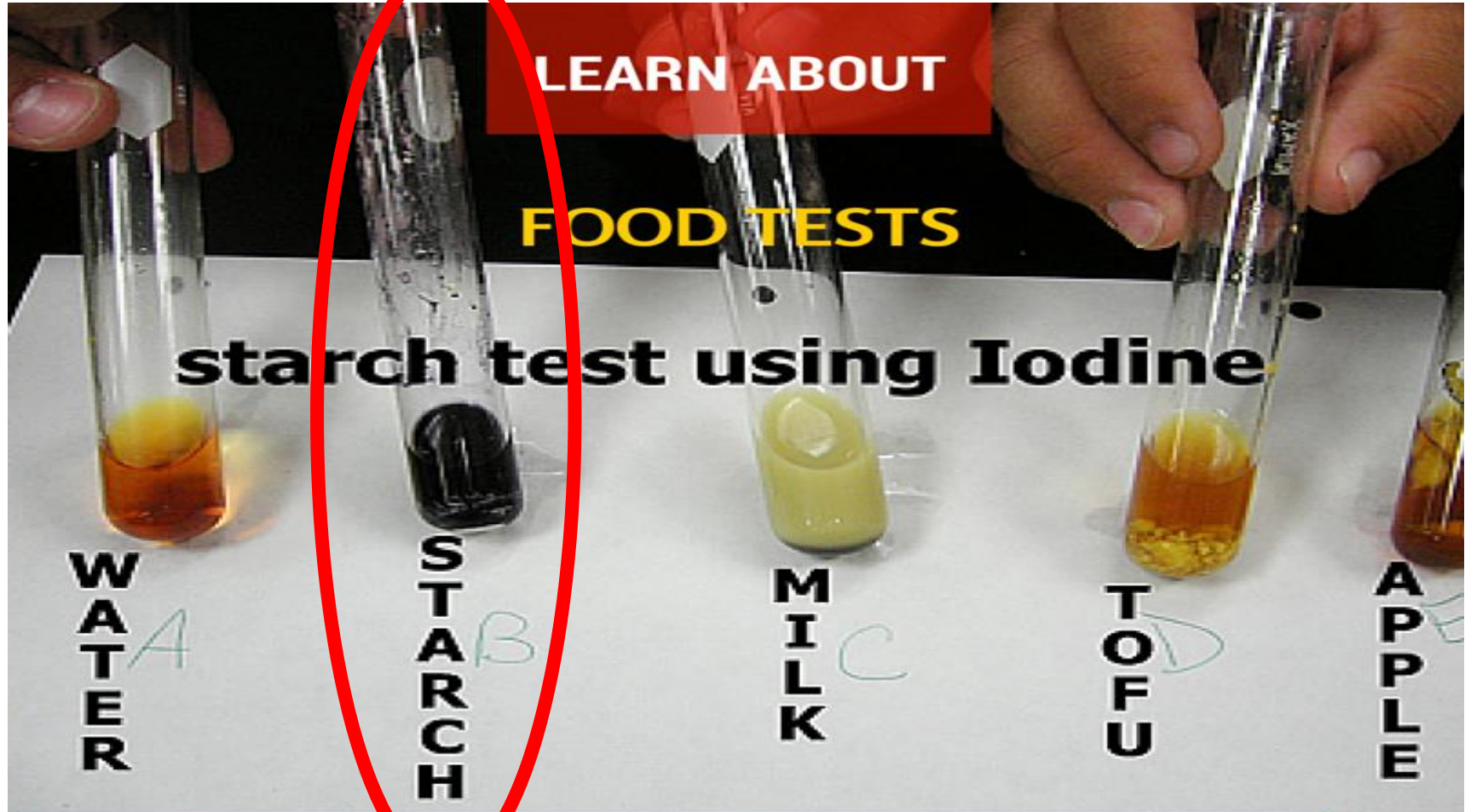


Distilled water: contains only water molecules



Here is a food test using the reagent iodine. What do you notice has happened?

Which test tube shows a positive result?



Brilliant Biology Student

Watch the experiments to see how food tests are done. Complete your table.

Watch
+ | listen

learn

Nutrient testing	Testing with		Result (change In colour)	
Starch	Benedicts		Purple	
	Iodine		Black	
	Ethanol		Yellow	
	Biuret		Emulsion	
Protein	Benedicts		Purple	
	Iodine		Black	
	Ethanol		Yellow	
	Biuret		Emulsion	
Sugar	Benedicts		Purple	
	Iodine		Black	
	Ethanol		Yellow	
	Biuret		Emulsion	
Fats and Lipids	Benedicts		Purple	
	Iodine		Black	
	Ethanol		Yellow	
	Biuret		Emulsion	

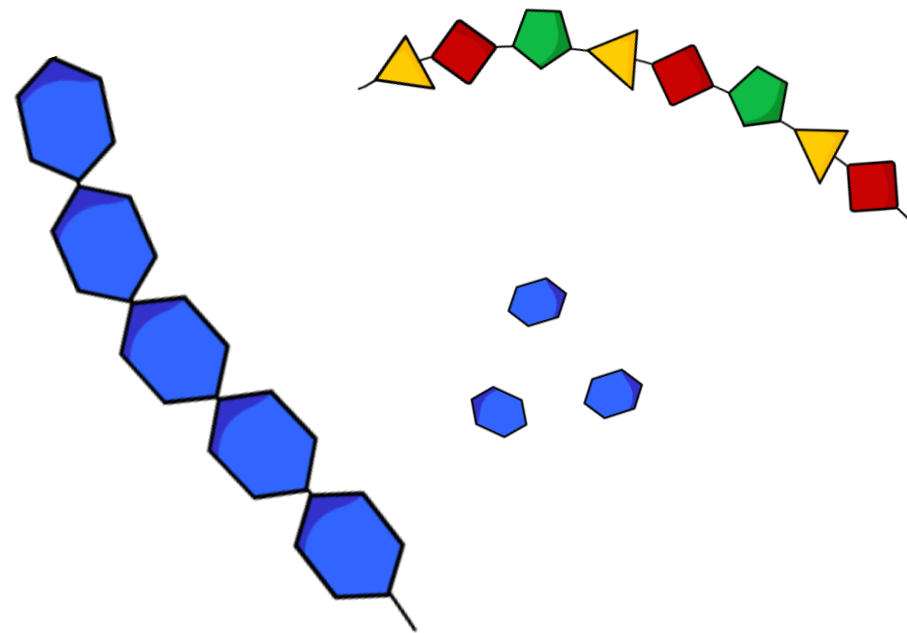
Nutrient testing	Testing with	Result (change In colour)
Starch	Benedicts Iodine Ethanol Biuret	
Protein	Benedicts Iodine Ethanol Biuret	
Sugar	Benedicts Iodine Ethanol Biuret	
Fats and Lipids	Benedicts Iodine Ethanol Biuret	

Review

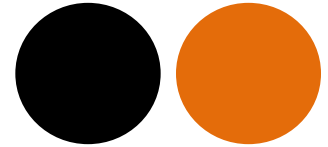
			(change In colour)	
Starch	Benedicts Iodine Ethanol Biuret	 	Purple Black Yellow Emulsion	
Protein	Benedicts Iodine Ethanol Biuret	 	Purple Black Yellow Emulsion	
Sugar	Benedicts Iodine Ethanol Biuret	 	Purple Black Yellow Emulsion	
Fats and Lipids	Benedicts Iodine Ethanol Biuret	 	Purple Black Yellow Emulsion	

Experiment

"Identify the nutrients in a food using reagents."



Can we match the variable to the pictures?



Colour change

INDEPENDENT VARIABLE

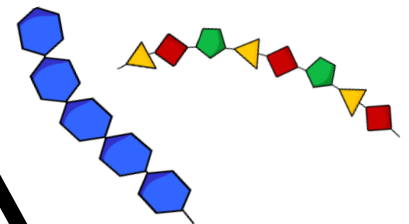


What I **CHANGE**



DEPENDENT VARIABLE

What I **OBSERVE**



Food type

Experiment 1: Testing for starch

1. Add 5cm³ of each food solution into each test tube making a note of which solution is in which test tube
2. Add a few drops of iodine solution to each test tube.
3. Observe and note the solution with a positive result = black

Experiment 3: Testing for sugars

1. Add 5cm³ of each food solution into each test tube making a note of which solution is in which tube
2. Add an equal volume of benedicts solution to each test tube.
3. Fill a beaker with hot water from a kettle
4. Add test tubes to the beaker.
Wait for a positive result = Yellow

Experiment 2: Testing for protein

1. Add 5cm³ of each food solution into each test tube making a note of which solution is in which test tube
2. Add an equal volume of biuret solution to each test tube.
3. Observe and note the solution with a positive result = purple

Experiment 4: Testing for fats

1. Add 5cm³ of each food solution into the each test tube making a note of which solution is in which test tube
2. Add 2cm³ of ethanol solution to each test tube and shake vigorously
3. After shaking add 2cm³ of water to each test tube. Observe positive result = white cloudiness

Food	Colour at the end of the test.			
	Iodine test	Benedict's test	Biuret test	Emulsion test
A – Biscuit				
B - Cheese				
C - Carrot				
D – Potato				

Review

Food	Colour at the end of the test.			
	Iodine test	Benedict's test	Biuret test	Emulsion test
A – Milk	orange	yellow	Purple	Emulsion
B - starch	black	blue	Blue	Clear
C - egg	orange	white	purple	clear
D – oil	orange	Yellow-white	Blue	Emulsion

LOb: Understand the chemistry of food and how they can be tested.

Keywords: Sugar, protein, lipids, nutrients, food test, starch, benedicts, iodine, biuret

Learning Outcomes:

- ✓ Recall the structure and role of the main nutrients
- ✓ Describe how to test foods for starch, sugar, lipids and protein
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Did you know... iodine is in group 7 of the periodic table. It has a lot of scientific uses. 1) Mixed with alcohol it can disinfect wounds 2) It is mixed with salt sometimes to prevent thyroid disease 3) Used as a catalyst.

keyfacts



Important **nutrients** include, **carbohydrates**, **proteins** and **lipids**.



Carbohydrates → provide energy → iodine reagent → **Black**



Protein → growth and development → biuret → **purple**



Sugars → source of fuel → benedict's reagent → **yellow**



Lipids → Efficient fuel → ethanol → **emulsion cloudy white-clear**

WRITE

2

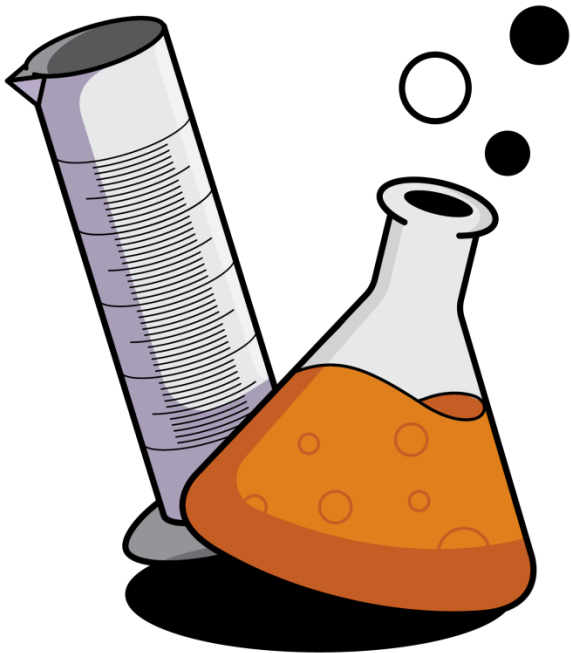
THINGS



you

LEARNT

[ABOUT]



The
Experiment

Plenary: Exam question

A student wants to test if there is any **difference between cola** and **diet cola**.

He wants to perform a scientific food test.

Write a plan for his test **suggesting the chemical** to use and **what the positive result will be** and **why**.

HELP

Benedicts solution

Warm Bath

Solution turns Orange if it has sugar

Diet coke?

QWC question (6 marks).

6

Example answers:

- Sensible equipment including test tube, Benedict's solution.
- Take a sample of coke, add Benedict's solution, heat it.
- Predict stated with reason.
- No colour change (will not turn yellow/red on heating).



Liam wants a test to find out if diet coke contains any sugar.

Write a plan for Liam to use to test for sugar and predict the result he should get, with a reason. (6 marks)

In this question you get marks for how well your answer is written.

You will get marks for:

- spelling
- grammar
- organising your ideas and information clearly
- using key scientific words.

HELP

Benedicts solution

Warm Bath

Solution turns Orange if it has sugar

Diet Lemonade?

QWC question (6 marks).

6

Example answers:

- Sensible equipment including test tube, Benedict's solution.
- Take a sample of lemonade, add Benedict's solution, heat it.
- Predict stated with reason.
- No colour change (will not turn yellow/red on heating).